LAPORAN
WORKSHOP DEVELOPER OPERATIONAL
MINGGU KE- 6

Dosen Pengampuh :

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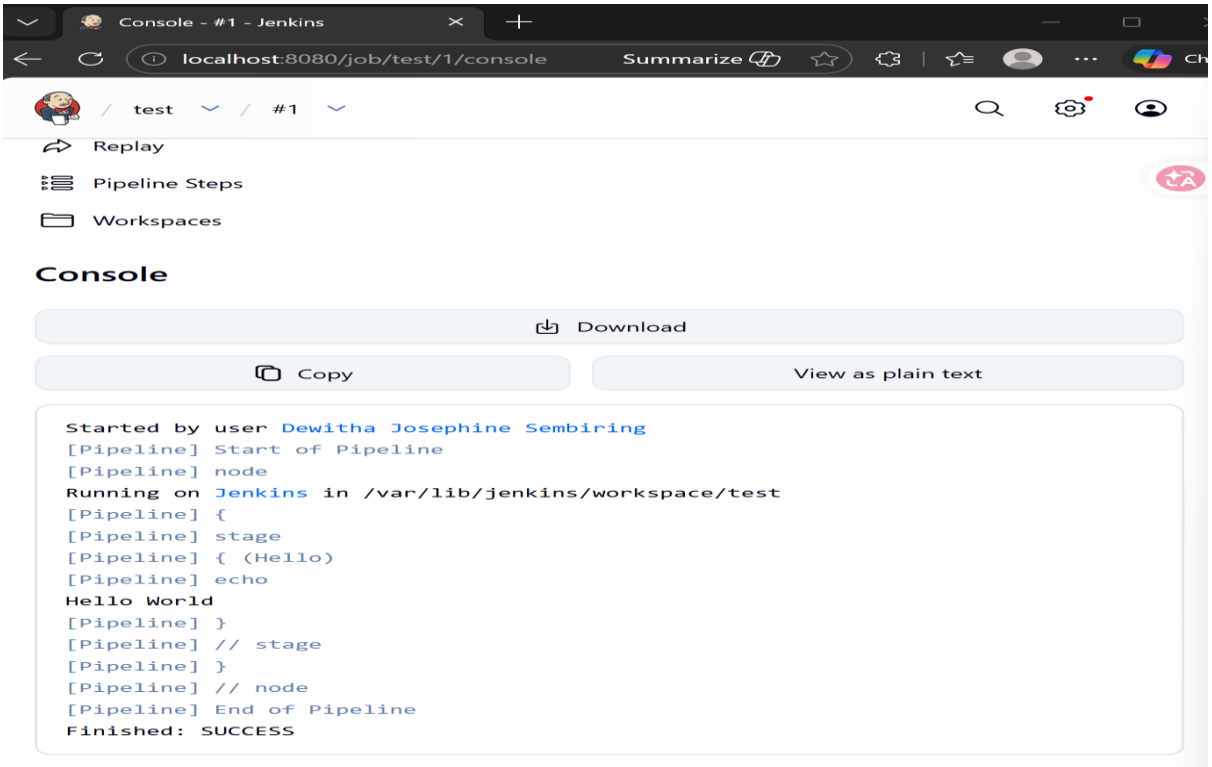
1. Jenkins Installation (WSL)

```
libra@Dewitha: /mnt/c/Users/

libra@Dewitha:~$ sudo apt install openjdk-17-jdk -y
[sudo] password for libra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java
  fonts-dejavu-extra java-common libasound2-data libasound2t64
  libatk-wrapper-java libatk-wrapper-java-jni libgif7 libice-dev
  libice6 libnspr4 libnss3 libpcsclite1 libpthread-stubs0-dev
  libsm-dev libsm6 libx11-dev libxau-dev libxaw7 libxcb-shape0
  libxcb1-dev libxdmcp-dev libxft2 libxkbfile1 libxmu6 libxpm4
  libxt-dev libxt6t64 libxv1 libxxf86dga1 openjdk-17-jdk-headless
  openjdk-17-jre openjdk-17-jre-headless x11-utils x11proto-dev
  xorg-sgml-doctools xtrans-dev
Suggested packages:
  default-jre alsa-utils libasound2-plugins libice-doc pcscd libsm-doc
  libx11-doc libxcb-doc libxt-doc openjdk-17-demo openjdk-17-source
  visualvm libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho
  fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic mesa-utils
Recommended packages:
  luit
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java
  fonts-dejavu-extra java-common libasound2-data libasound2t64
  libatk-wrapper-java libatk-wrapper-java-jni libgif7 libice-dev
```

```
libra@Dewitha:~$ java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, s
haring)
```

```
libra@Dewitha:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-03-10 13:32:10--  https://pkg.jenkins.io/debian-stable/jenkins.io
-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.2.133, 151.101.66.1
33, 151.101.194.133, ...
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.2.133|:443... conn
ected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings 100%[==============>]   1.64K  --.-KB/s    in 0s

2026-03-10 13:32:11 (34.4 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc
' saved [1680/1680]

libra@Dewitha:~$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.
asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
libra@Dewitha:~$ sudo apt install jenkins -y
sudo systemctl start jenkins
sudo systemctl enable jenkins
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package jenkins is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
```

```
libra@Dewitha:~$ sudo service jenkins start
 * Starting Jenkins Automation Server jenkins
Setting up max open files limit to 8192
                                                                    [ OK ]
libra@Dewitha:~$ sudo service jenkins status
 * jenkins is running
libra@Dewitha:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
aed91429d2c346c5b01c0aa332b33746
libra@Dewitha:~$ cd /mnt/c/Users/libra/Jenkins
libra@Dewitha:/mnt/c/Users/libra/Jenkins$ mkdir website
cd website

mkdir laravel_jenkins
cd laravel_jenkins
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins$ compos
er -v
   ______
  / ____/___  ____ ___  ____  ____  ________  _____
 / /   / __ \/ __ `__ \/ __ \/ __ \/ ___/ _ \/ ___/
/ /___/ /_/ / / / / / / /_/ / /_/ (__  )  __/ /
\____/\____/_/ /_/ /_/ .___/\____/____/\___/_/
                    /_/
Composer version 2.7.1 2024-02-09 15:26:28

Usage:
  command [options] [arguments]

Options:
  -h, --help                     Display help for the given command. Whe
n no command is given display help for the list command
  -q, --quiet                    Do not output any message
  -V, --version                  Display this application version
      --ansi|--no-ansi           Force (or disable --no-ansi) ANSI outpu
t
```



Pada tahap ini dilakukan instalasi Jenkins pada lingkungan Windows Subsystem for Linux. Sebelum menginstal Jenkins, terlebih dahulu dilakukan instalasi OpenJDK versi 17 karena Jenkins berjalan di atas Java Virtual Machine. Instalasi dilakukan

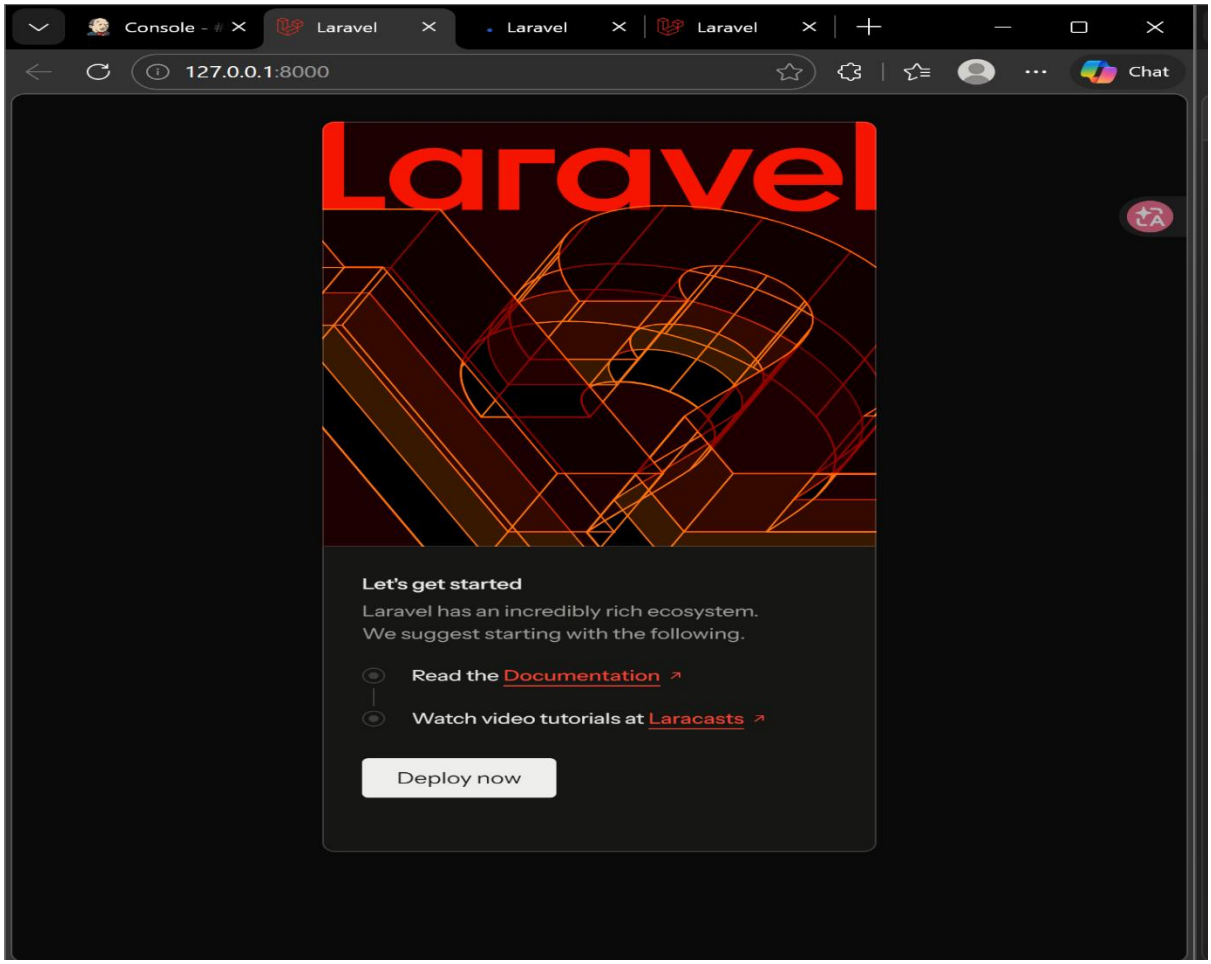
menggunakan perintah apt install openjdk-17-jdk, kemudian dilakukan pengecekan dengan java -version untuk memastikan Java telah berhasil terpasang.

Setelah itu repository Jenkins ditambahkan ke sistem Linux dan Jenkins diinstal menggunakan package manager. Layanan Jenkins kemudian dijalankan dan dapat diakses melalui browser pada alamat http://localhost:8080. Untuk login pertama kali, diperlukan initial admin password yang diambil dari terminal. Setelah masuk ke dashboard Jenkins, dilakukan instalasi plugin yang disarankan dan pengujian pipeline sederhana menggunakan script "Hello World". Jika proses build berhasil, maka pada Console Output akan muncul status Finished: SUCCESS, yang menandakan Jenkins telah berhasil berjalan dengan baik.

2. Laravel Project Setup

```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins$ compos
er create-project laravel/laravel laravel-ci
Creating a "laravel/laravel" project at "./laravel-ci"
Installing laravel/laravel (v12.11.2)
  - Installing laravel/laravel (v12.11.2): Extracting archive
Created project in /mnt/c/Users/libra/Jenkins/website/laravel_jenkins/la
ravel-ci
> @php -r "file_exists('.env') || copy('.env.example', '.env');"
Loading composer repositories with package information
Updating dependencies
Lock file operations: 111 installs, 0 updates, 0 removals
  - Locking brick/math (0.14.8)
  - Locking carbonphp/carbon-doctrine-types (3.2.0)
  - Locking dflydev/dot-access-data (v3.0.3)
  - Locking doctrine/inflector (2.1.0)
  - Locking doctrine/lexer (3.0.1)
  - Locking dragonmantank/cron-expression (v3.6.0)
  - Locking egulias/email-validator (4.0.4)
  - Locking fakerphp/faker (v1.24.1)
  - Locking filp/whoops (2.18.4)
  - Locking fruitcake/php-cors (v1.4.0)
  - Locking graham-campbell/result-type (v1.1.4)
  - Locking guzzlehttp/guzzle (7.10.0)
  - Locking guzzlehttp/promises (2.3.0)
  - Locking guzzlehttp/psr7 (2.8.0)
  - Locking guzzlehttp/uri-template (v1.0.5)
```

```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins$  cd la
ravel-ci
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ cp .env.example .env
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ php artisan key:generate

  INFO  Application key set successfully.

libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].

  Press Ctrl+C to stop the server

  2026-03-10 14:10:24 / ..................................... ~ 1m 20s
  2026-03-10 14:11:45 / ..................................... ~ 1m 31s
  2026-03-10 14:11:45 / ..................................... ~ 2m 4s
^Clibra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/larav
el-cisudo apt install php8.3-sqlite3 -y-y
[sudo] password for libra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  php8.3-sqlite3
0 upgraded, 1 newly installed, 0 to remove and 6 not upgraded.
```



Pada tahap ini dilakukan pembuatan proyek menggunakan framework Laravel. Pertama dibuat folder project pada direktori Documents, kemudian dibuat folder khusus untuk pengembangan proyek. Setelah itu dilakukan instalasi Laravel menggunakan Composer dengan perintah composer create-project laravel/laravel laravel-ci, yang akan mengunduh dan menyiapkan struktur dasar aplikasi Laravel secara otomatis.

Setelah instalasi selesai, dilakukan konfigurasi environment dengan menyalin file .env.example menjadi .env, kemudian menghasilkan application key menggunakan perintah php artisan key:generate. Terakhir, aplikasi dijalankan menggunakan php artisan serve sehingga Laravel dapat diakses melalui browser pada alamat http://127.0.0.1:8000 untuk memastikan bahwa proyek telah berhasil dibuat dan berjalan dengan baik.

3. CI/CD Using GitHub Actions

```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git init
hint: Using 'master' as the name for the initial branch. This default br
anch name
hint: is subject to change. To configure the initial branch name to use
in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this com
mand:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/libra/Jenkins/website/l
aravel_jenkins/laravel-ci/.git/
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git add .
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git commit -m "Initial Laravel CI project"
Author identity unknown
```

```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git config --global user.name "Dewitha"
git config --global user.email "dewitha.j@gmail.com"
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git config --global --list
user.name=Dewitha
user.email=dewitha.j@gmail.com
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git init
Reinitialized existing Git repository in /mnt/c/Users/libra/Jenkins/webs
ite/laravel_jenkins/laravel-ci/.git/
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git add .
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git commit -m "Initial Laravel CI project"
[master (root-commit) 0b0d8b3] Initial Laravel CI project
 56 files changed, 10749 insertions(+)
 create mode 100644 .editorconfig
 create mode 100644 .env.example
 create mode 100644 .gitattributes
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 app/Http/Controllers/Controller.php
 create mode 100644 app/Models/User.php
 create mode 100644 app/Providers/AppServiceProvider.php
 create mode 100644 artisan
 create mode 100644 bootstrap/app.php
 create mode 100644 bootstrap/cache/.gitignore
 create mode 100644 bootstrap/providers.php
 create mode 100644 composer.json
 create mode 100644 composer.lock
 create mode 100644 config/app.php
 create mode 100644 config/auth.php
 create mode 100644 config/cache.php
```

Files main laravel-ci / .github / workflows / laravel.yml

Dewitha Add GitHub Actions CI d7d86fe · 11 hours ago

33 lines (24 loc) · 606 Bytes

Code Blame Raw

```yaml
name: Laravel CI

on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]

jobs:
  laravel-tests:

    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Setup PHP
        uses: shivammathur/setup-php@v2
        with:
          php-version: 8.2

      - name: Install dependencies
        run: composer install --no-interaction --prefer-dist

      - name: Copy ENV
        run: cp .env.example .env
```

Add GitHub Actions CI · Dewitha Dewitha/laravel-ci

https://github.com/Dewitha/laravel-ci/action... Chat

Dewitha / laravel-ci

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Laravel CI

Add GitHub Actions CI #1

Summary

Triggered via push 13 minutes ago

Dewitha pushed d7d86fe main

Status	Total duration	Artifacts
Success	29s	–

laravel.yml

on: push

laravel-tests 25s

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Pada tahap ini dilakukan penerapan Continuous Integration / Continuous Deployment (CI/CD) menggunakan GitHub Actions pada repository proyek yang berada di GitHub. Pertama, proyek Laravel yang telah dibuat sebelumnya di-push ke repository GitHub. Selanjutnya dibuat folder .github/workflows yang berfungsi sebagai tempat penyimpanan file konfigurasi workflow otomatis.
Di dalam folder tersebut dibuat file laravel.yml yang berisi konfigurasi pipeline CI. Workflow ini akan berjalan secara otomatis setiap kali terjadi push atau pull request pada branch main. Proses yang dijalankan meliputi pengambilan source code dari repository, setup lingkungan PHP, instalasi dependensi menggunakan Composer, pembuatan file environment, pembuatan application key, serta menjalankan pengujian aplikasi dengan perintah php artisan test. Dengan adanya workflow ini, proses pengujian aplikasi dapat berjalan secara otomatis setiap kali terjadi perubahan kode pada repository.

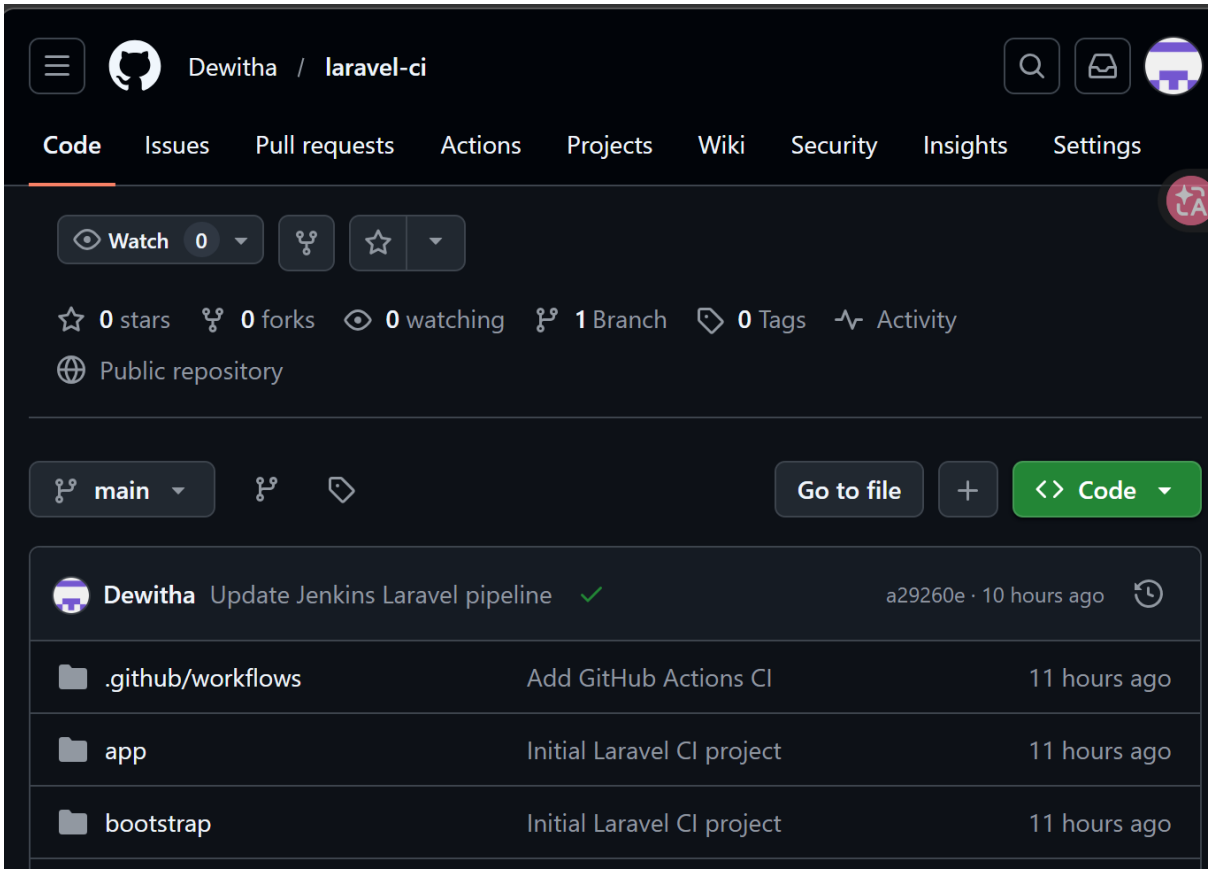
4. CI/CD Using Jenkins

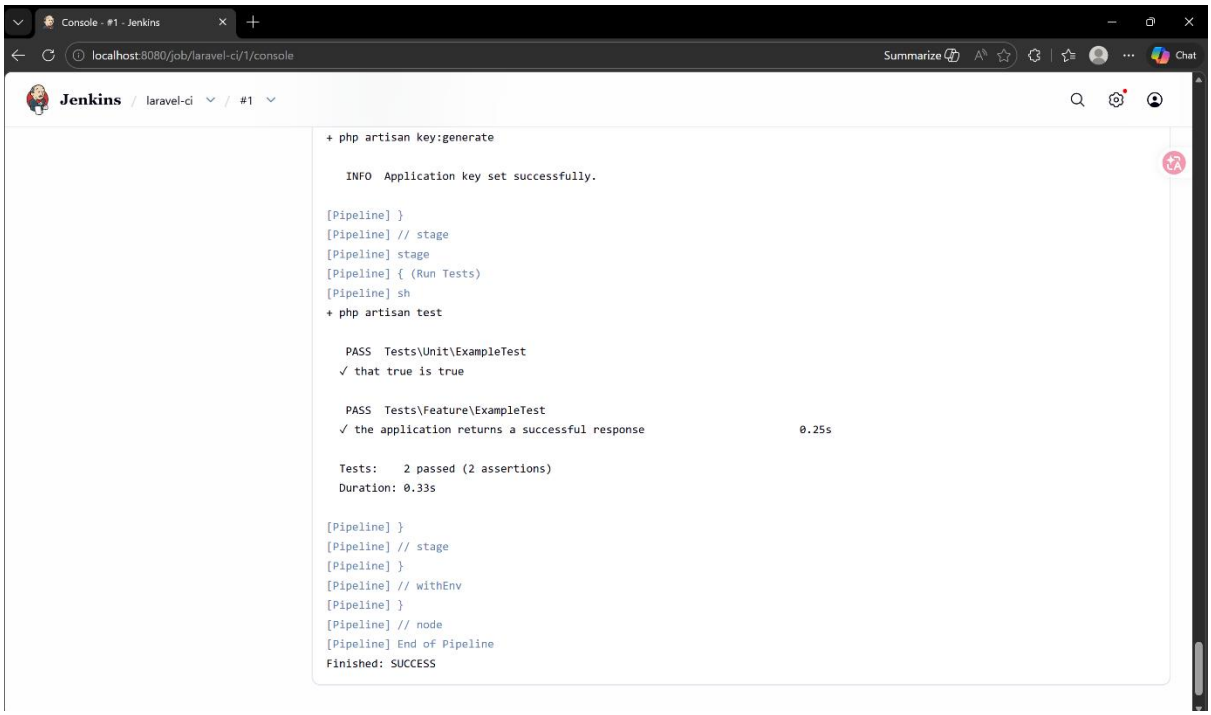
```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ sudo apt update
sudo apt install php php-cli php-mbstring php-xml php-curl php-zip unzip
 curl git -y
[sudo] password for libra:
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Ign:2 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:3 https://pkg.jenkins.io/debian-stable binary/ Release
Get:4 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB
]
Hit:7 http://archive.ubuntu.com/ubuntu noble InRelease
Get:8 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Compon
ents [21.5 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 C
omponents [74.2 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64
 Components [212 B]
Get:12 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64
 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 k
B]
```

```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ cd /mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel-ci
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ nano Jenkinsfile
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git add .
git commit -m "Add Jenkins pipeline"
git push
[main bca8889] Add Jenkins pipeline
 1 file changed, 15 insertions(+)
 create mode 100644 Jenkinsfile
Username for 'https://github.com': Dewitha
Password for 'https://Dewitha@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 382 bytes | 11.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Dewitha/laravel-ci.git
   d7d86fe..bca8889  main -> main
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ nano Jenkinsfile
```

```
libra@Dewitha: /mnt/c/Users/
  GNU nano 7.2                    Jenkinsfile *
pipeline {
    agent any

    stages {

        stage('Check Tools') {
            steps {
                sh 'git --version'
                sh 'php -v'
                sh 'composer -V'
            }
        }

    }
}
```

```
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ nano Jenkinsfile
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$ git add .
git commit -m "Update Jenkins Laravel pipeline"
git push
[main a29260e] Update Jenkins Laravel pipeline
 1 file changed, 21 insertions(+), 4 deletions(-)
Username for 'https://github.com': Dewitha
Password for 'https://Dewitha@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 473 bytes | 10.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Dewitha/laravel-ci.git
   bca8889..a29260e  main -> main
libra@Dewitha:/mnt/c/Users/libra/Jenkins/website/laravel_jenkins/laravel
-ci$
```





Pada tahap ini dilakukan penerapan CI/CD menggunakan Jenkins. Pertama dilakukan instalasi beberapa plugin yang diperlukan melalui menu Manage Jenkins → Plugins, seperti Git Plugin, GitHub Integration, Credentials Binding, Pipeline, dan Pipeline Stage View. Selain itu juga dilakukan instalasi tools yang dibutuhkan untuk menjalankan proyek Laravel, yaitu PHP dan Composer. Setelah instalasi selesai,

dilakukan pengujian pada pipeline Jenkins untuk memastikan bahwa tools seperti Git, PHP, dan Composer dapat berjalan dengan baik.

Selanjutnya dibuat file Jenkinsfile pada root project yang berisi konfigurasi pipeline untuk menjalankan proses otomatis, seperti mengambil source code dari GitHub, menginstal dependensi menggunakan Composer, melakukan konfigurasi Laravel, serta menjalankan pengujian aplikasi. Pipeline kemudian dikonfigurasi di Jenkins dengan memilih Pipeline script from SCM dan menghubungkannya dengan repository GitHub. Untuk mengakses repository, ditambahkan kredensial berupa GitHub username dan Personal Access Token. Setelah konfigurasi selesai, pipeline dijalankan menggunakan Build Now, dan jika seluruh proses berhasil maka akan muncul status Finished: SUCCESS.